

Whitley Thickness Is Not Bidual-Invariant

Literature status for the question in arXiv:1307.4385

Automated mathematics run `fa_banach_001`, lane 1

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Status: literature already answered. The question in Castillo–Papini–Simões is answered negatively by Abrahamsen–Langemets–Lima–Nygaard, who explicitly identify their example as answering that question.

Source question

J. Castillo, P. L. Papini, and M. Simões, *Thick coverings for the unit ball of a Banach space*, arXiv:1307.4385 [1], PDF page 7, asks:

“A trying question posed in [3] is whether $T(X) = T(X^{**})$.”

Here $T(X)$ is Whitley’s thickness constant, equivalently for an infinite-dimensional Banach space the infimum of radii r for which finitely many radius- r balls centered on S_X cover B_X .

Later answer

T. A. Abrahamsen, J. Langemets, V. Lima, and O. Nygaard, *On thickness and thinness of Banach spaces*, arXiv:1402.0996 [2], PDF page 8, first note the general inequality

$$T(X^{**}) \leq T(X),$$

and then state that it can be strict. Their explicit example is

$$T(C[0, 1]) = 2, \quad T(C[0, 1]^{**}) = 1.$$

They immediately add: “Note that this example answers a question in [7] whether we always have $T(X) = T(X^{**})$.” Their reference [7] is precisely the source paper above.

Thus the answer to the source question is **no**. The supporting authors were explicitly aware that their example answered the same question. This is a direct literature answer rather than an agent-identified implication.

Scope and search evidence

The counterexample settles the unrestricted equality question completely. It does not assert a classification of spaces for which equality holds. The run’s four lightweight indexes had no entry for arXiv:1307.4385. Local exact-phrase search located the explicit answer in arXiv:1402.0996 and an independent 2019 paper which cites the same 2015 Houston Journal article as the negative solution. A bounded web/arXiv search confirmed the supporting paper’s metadata.

References

- [1] J. Castillo, P. L. Papini, and M. Simões, *Thick coverings for the unit ball of a Banach space*, Houston J. Math. **41** (2015), 177–189; arXiv:1307.4385.
- [2] T. A. Abrahamsen, J. Langemets, V. Lima, and O. Nygaard, *On thickness and thinness of Banach spaces*, Houston J. Math. **41** (2015), 97–111; arXiv:1402.0996.